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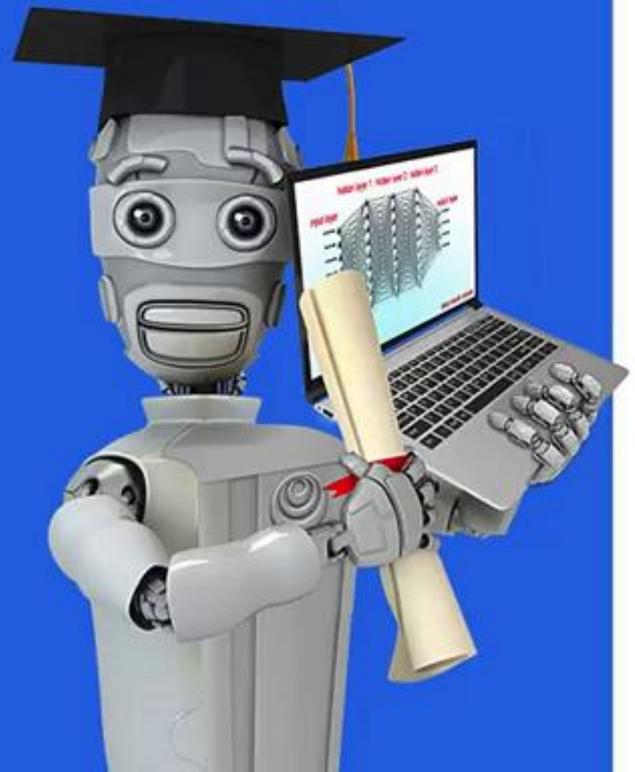


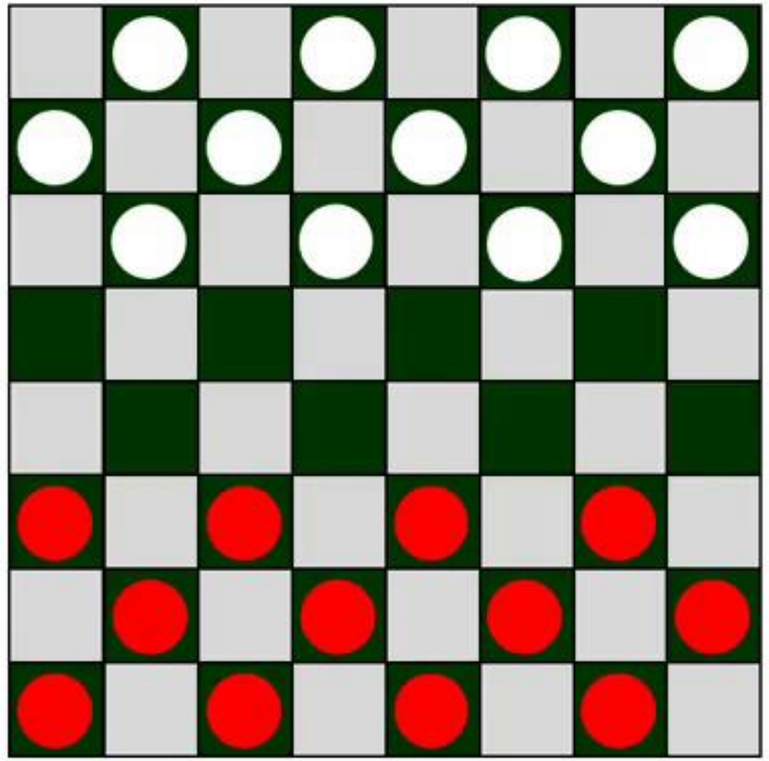
# Machine Learning Overview

## What is Machine Learning?

# Machine learning

"Field of study that gives computers the ability to learn without being explicitly programmed."

Arthur Samuel (1959)



# Question

If the checkers program had been allowed to play only ten games (instead of tens of thousands) against itself, a much smaller number of games, how would this have affected its performance?

- Would have made it better
- Would have made it worse

# Machine learning algorithms

rapid advancements

used most in real-world applications

- Supervised learning ← course 1, 2

- Unsupervised learning ←
- Recommender systems
- Reinforcement learning

course 3

Practical advice for applying learning algorithms

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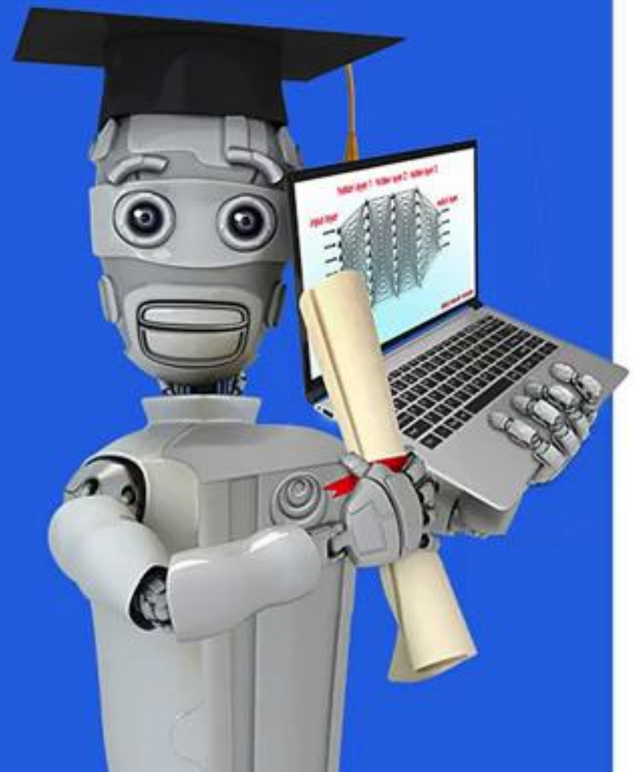


# Machine Learning Overview

## Supervised Learning Part 1

# Supervised learning

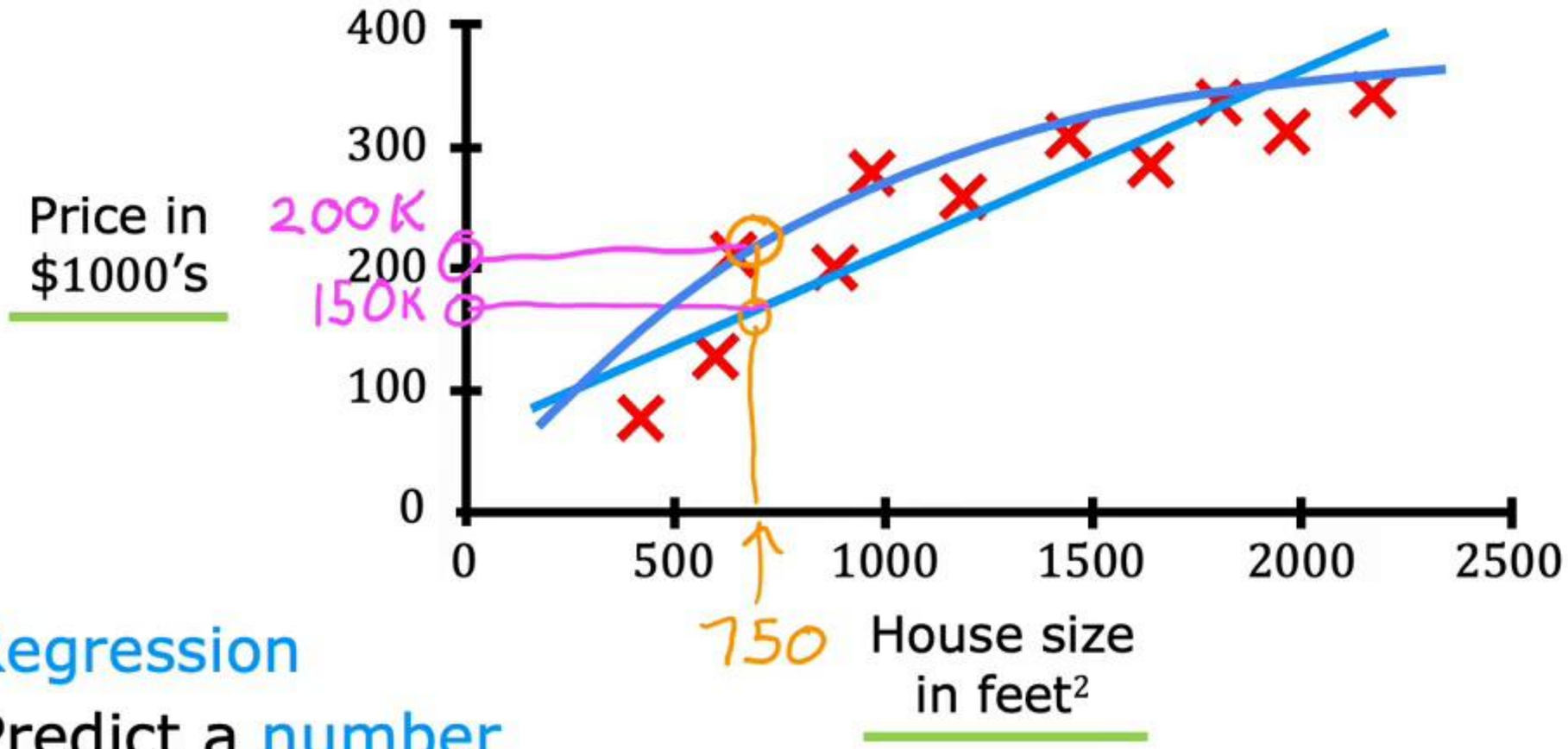
$x$ → $y$

input → output label

Learns from being given "right answers"

| Input (X) | Output (Y) | Application |
|---|---|---|
| email | spam? (0/1) | spam filtering |
| audio | text transcripts | speech recognition |
| English | Spanish | machine translation |
| ad, user info | click? (0/1) | online advertising |
| image, radar info | position of other cars | self-driving car |
| image of phone | defect? (0/1) | visual inspectic |

# Regression: Housing price prediction




Regression

Predict a number

infinitely many possible outputs

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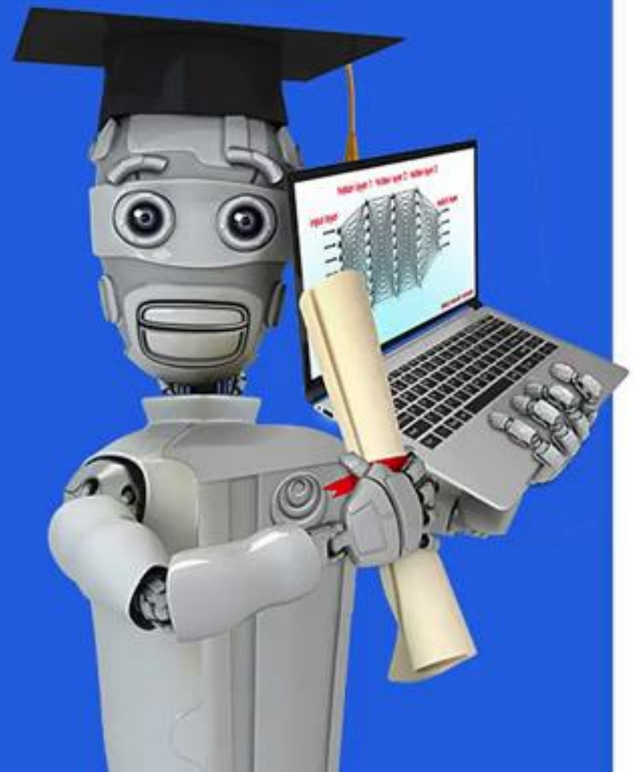
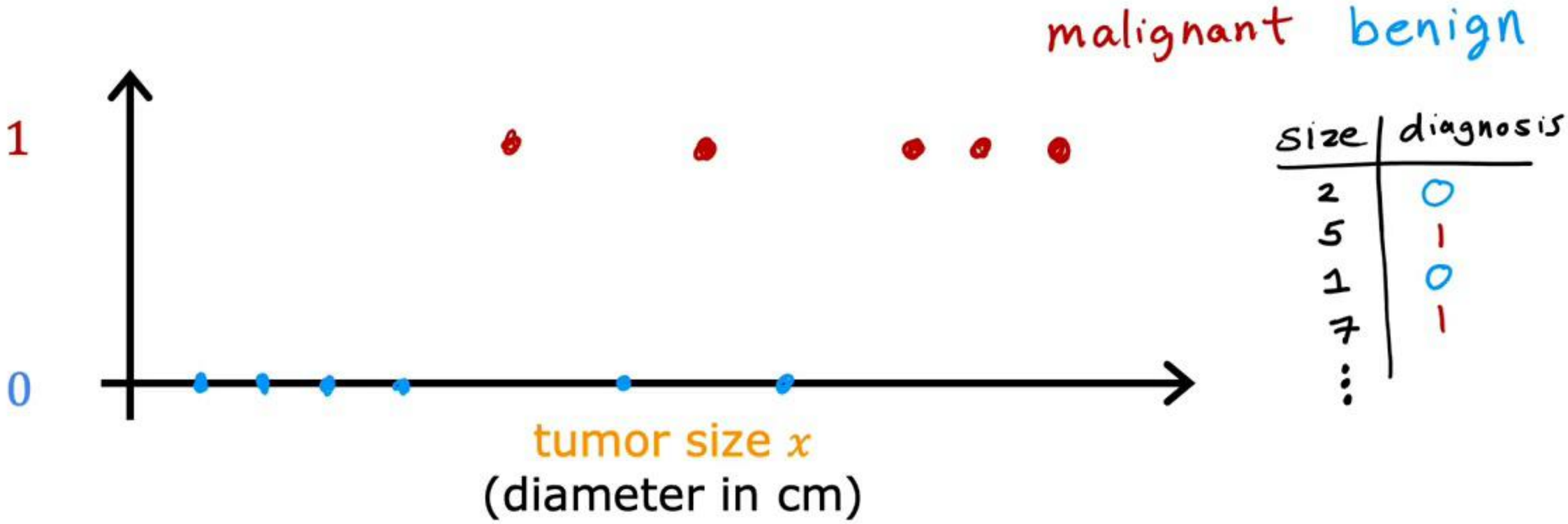


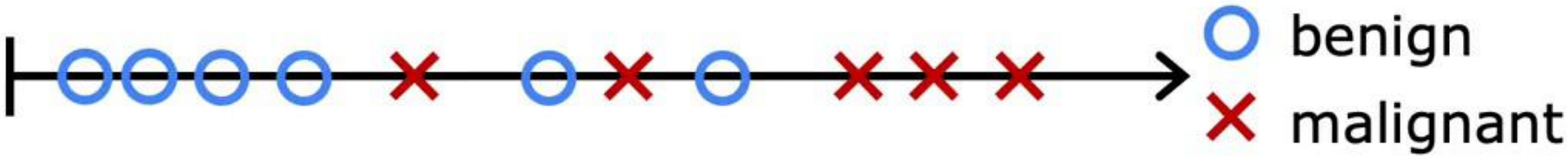
# Machine Learning Overview

## Supervised Learning Part 2

# Classification: Breast cancer detection




| size | diagnosis |
|---|---|
| 2 | 0 |
| 5 | 1 |
| 1 | 0 |
| 7 | 1 |
| ⋮ | |



# Classification: Breast cancer detection

- ○ benign
- × malignant type1
- △ malignant type 2

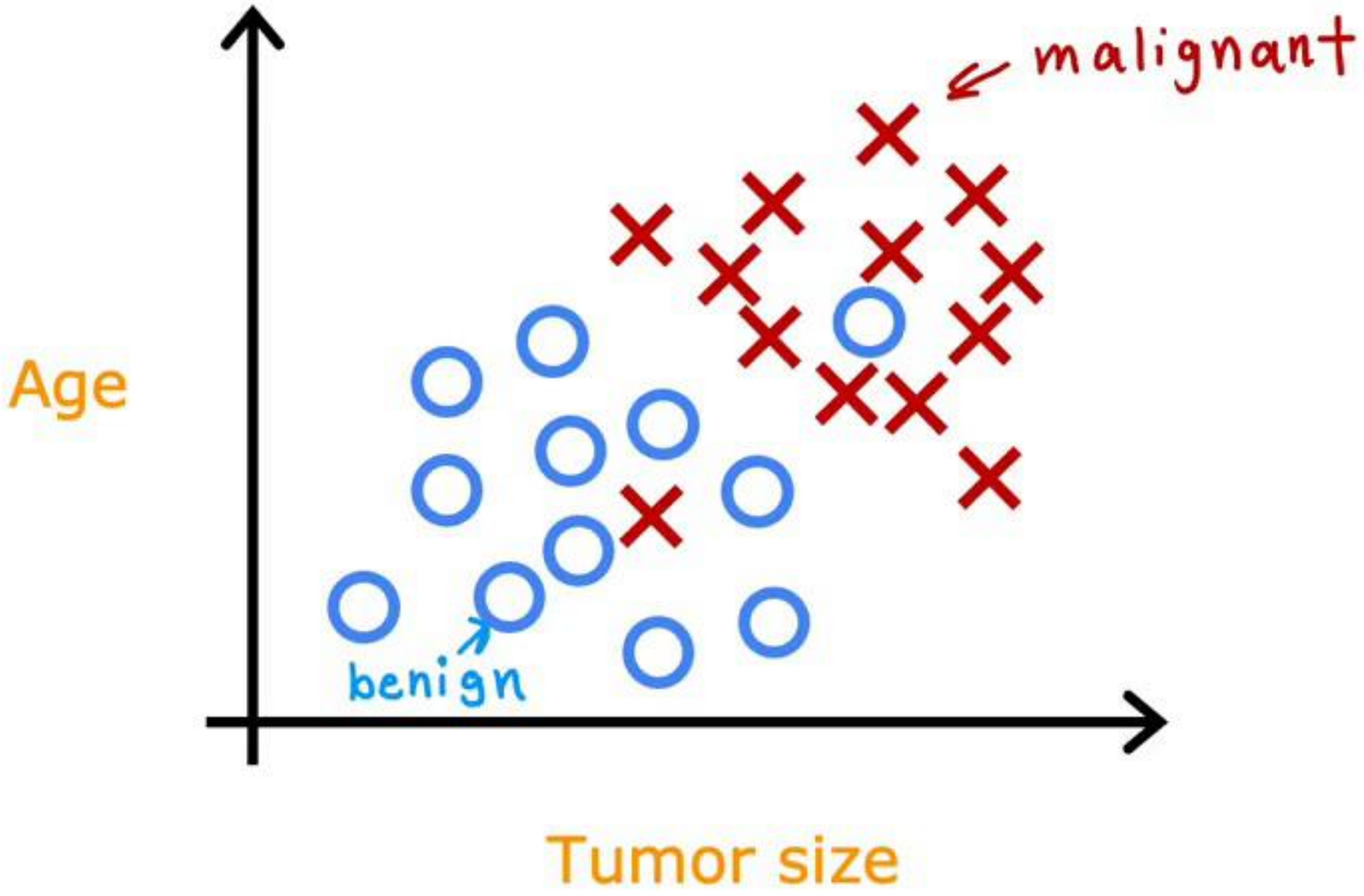
class category

Classification

predict categories cat dog benign malignant 0, 1, 2

small number of possible outputs

# Two or more inputs



# Supervised learning

Learns from being given "right answers"

**Regression**

Predict a number

infinitely many possible outputs

**Classification**

predict categories

small number of pos

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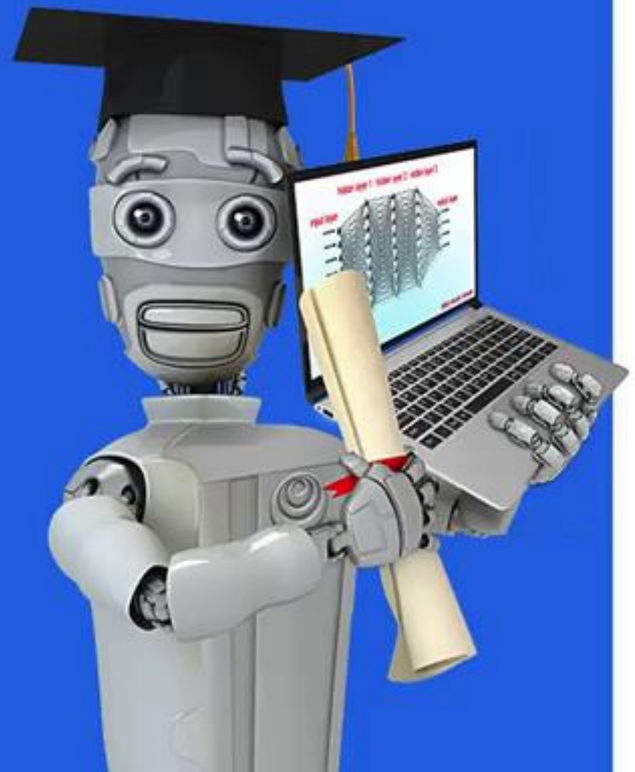
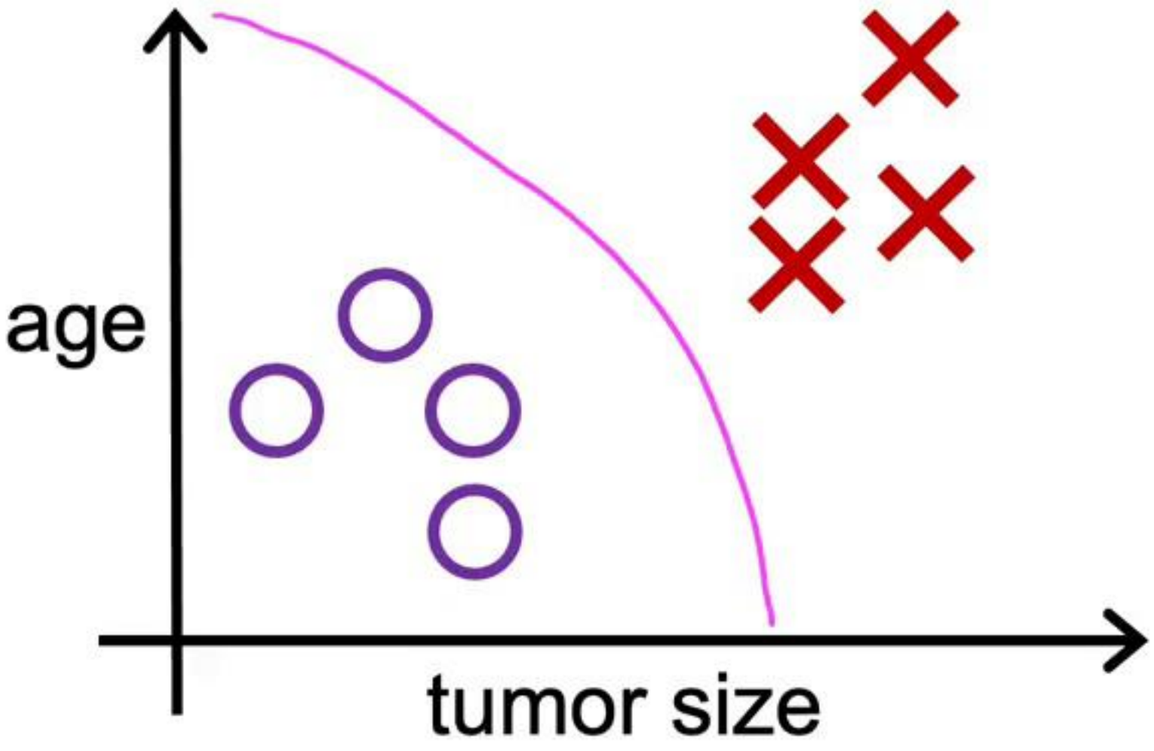


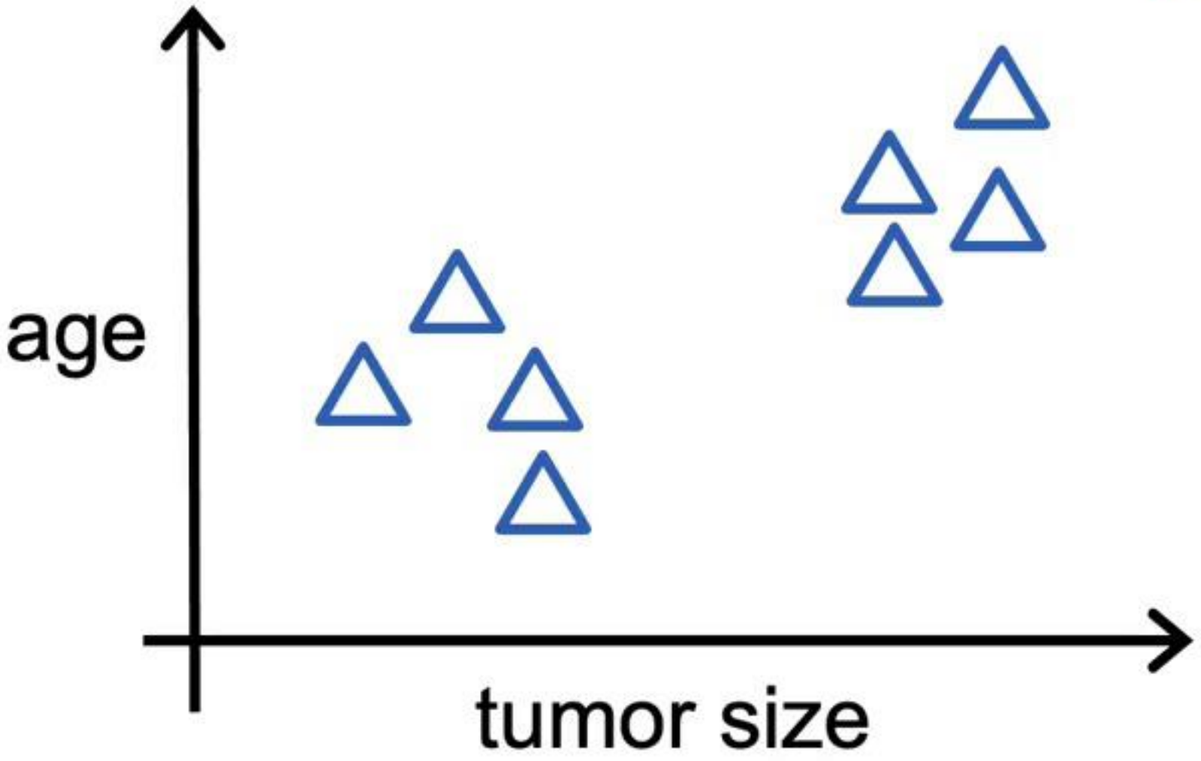
# Machine Learning Overview

## Unsupervised Learning Part 1

Supervised learning
Learn from data labeled with the "right answers"

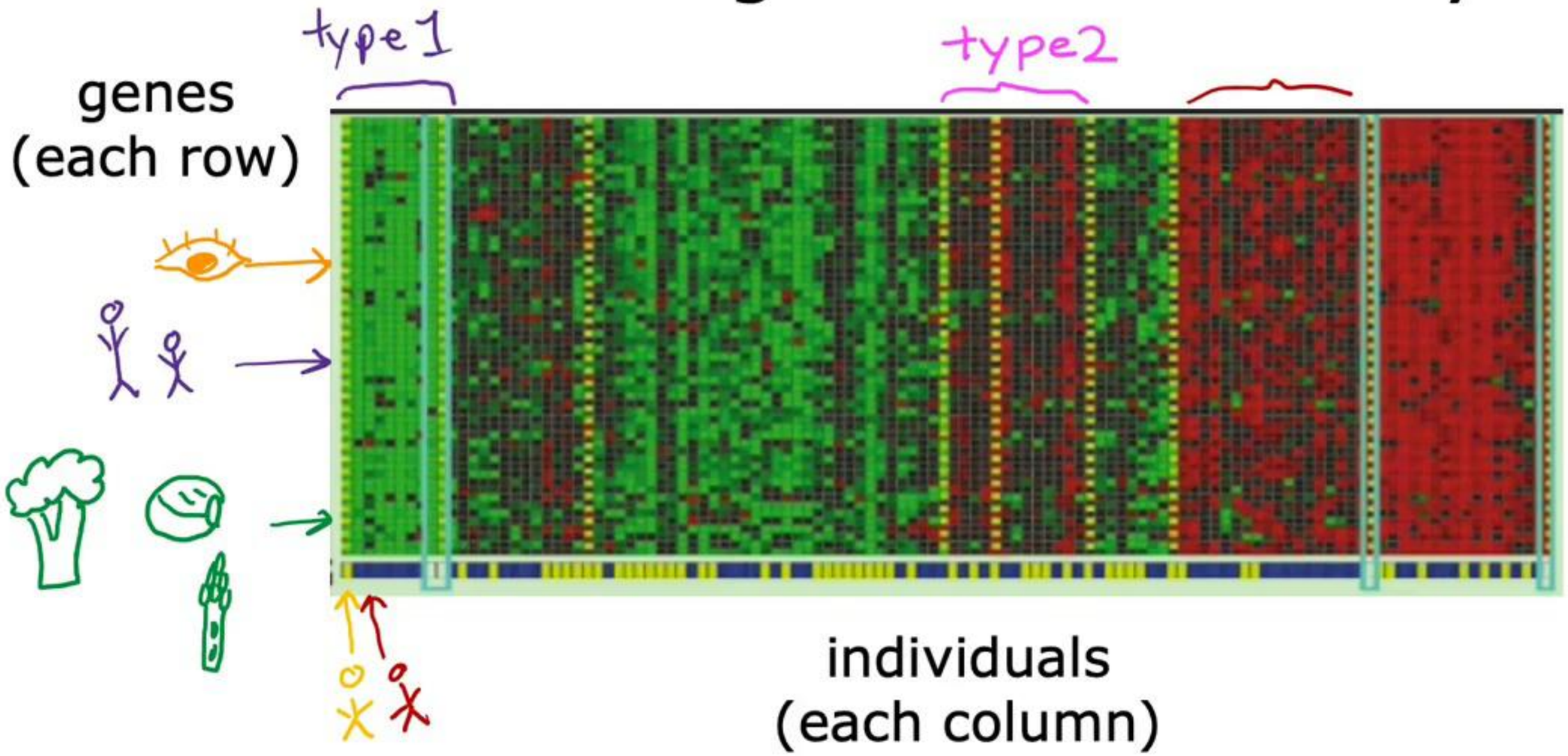



Unsupervised learning
Find something interesting in unlabeled data.

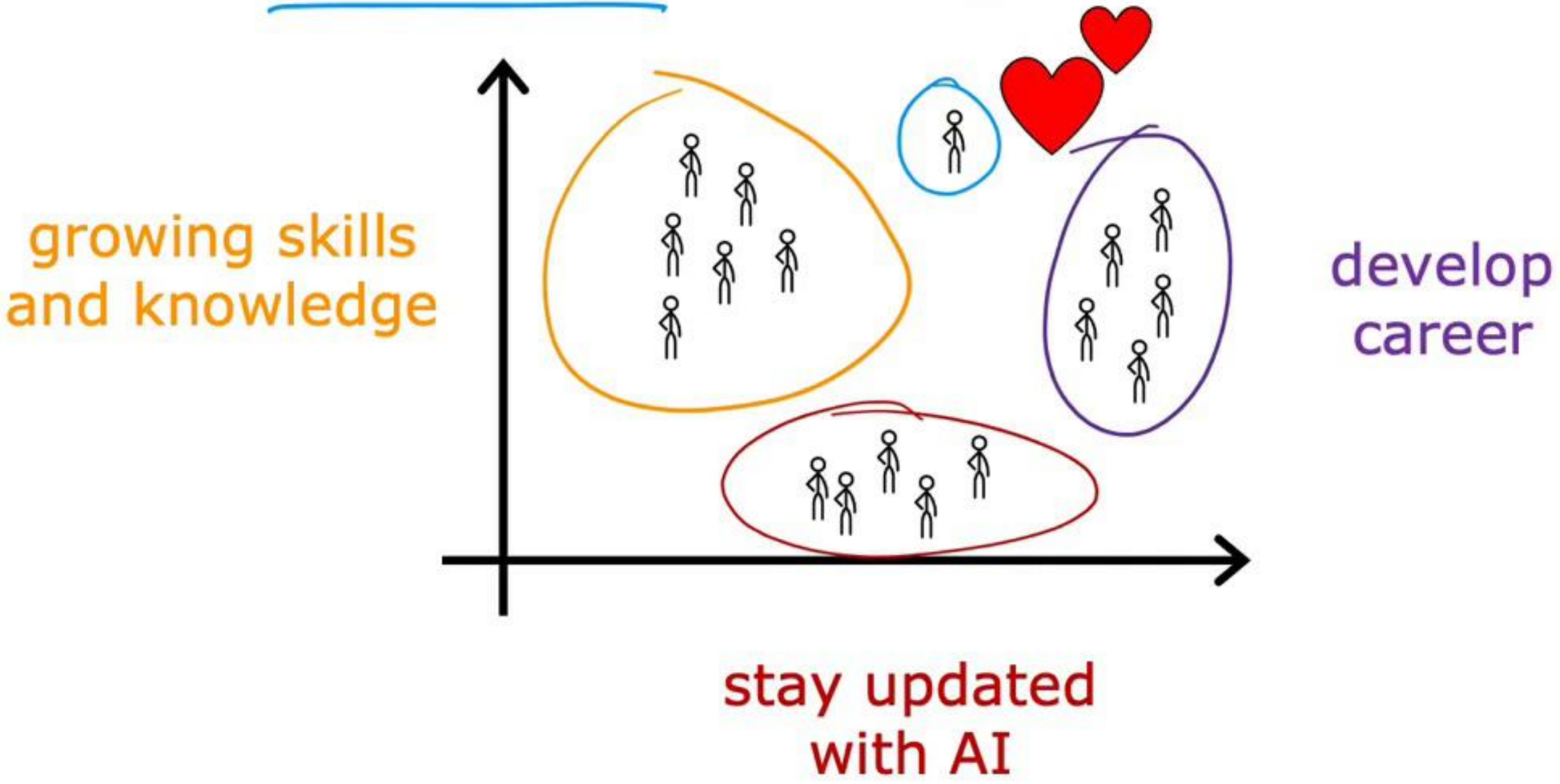


# Clustering: Google news

# Clustering: DNA microarray



# Clustering: Grouping customers



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# Machine Learning Overview

## Unsupervised Learning Part 2

# Unsupervised learning

Data only comes with inputs $x$, but not output labels $y$.
Algorithm has to find structure in the data.

Clustering
Group similar data points together.

Dimensionality reduction
Compress data using fewer numbers.

Anomaly detection
Find unusual data points.

# Question

Of the following examples, which would you address using an unsupervised learning algorithm?

- ☐ Given email labeled as spam/not spam, learn a spam filter.
- ☐ Given a set of news articles found on the web, group them into sets of articles about the same story.
- ☐ Given a database of customer data, automatically discover market segments and group customers into different market segments.
- ☐ Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not

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# Machine Learning Overview

## Jupyter Notebooks

Lab 1 Python and Jupyter Note

coursera.org/learn/mls-c1-supervised-ml-regression-classification/ungradedLab/fD1DQ/lab-1-python-and-jupyter-notebooks/lab?path=%2Fnotebooks%2FOptional%2520Lab%2520-%2520Brief%2520Introduction%2520to%2520Python%2520a...

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jupyter Optional Lab - Brief Introduction to Python and Jupyter Notebooks Last Checkpoint: 34 minutes ago (autosaved)

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Run Markdown

# Optional Lab: Brief Introduction to Python and Jupyter Notebooks

Welcome to the first optional lab! Optional labs are available to:

- provide information - like this notebook
- reinforce lecture material with hands-on examples
- provide working examples of routines used in the graded labs

## Goals

In this lab, you will:

- Get a brief introduction to Jupyter notebooks
- Take a tour of Jupyter notebooks
- Learn the difference between markdown cells and code cells
- Practice some basic python

The easiest way to become familiar with Jupyter notebooks is to take the tour available above in the Help menu:

Cell Kernel Widgets Help

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New Notebook
Open...
Make a Copy...
Save as...
Rename...
Save and Checkpoint
Revert to Checkpoint
Print Preview
Download as
Trust Notebook
Close and Halt

Run Markdown

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abs.coursera.org/notebooks/Optional Lab Brief Introduction to Python and Jupyter Notebooks.ipynb#

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Run

Interrupt
Restart
Restart & Clear Output
Restart & Run All
Reconnect
Shutdown

Change kernel

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qisxrs.labs.coursera.org/notebooks/Optional Lab Brief Introduction to Python and Jupyter Notebooks.ipynb#

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Jupyter notebooks have two types of cells that are used in this course. Cells such as this which contain documentation called `Markdown Cells` . The name is derived from the simple formatting language used in the cells. You will not be required to produce markdown cells. Its useful to understand the `cell pulldown` shown in graphic below. Occasionally, a cell will end up in the wrong mode and you may need to restore it to the right state:

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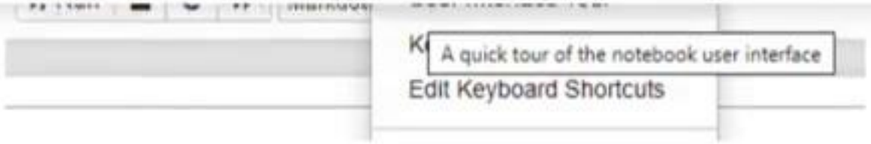
Run Markdown

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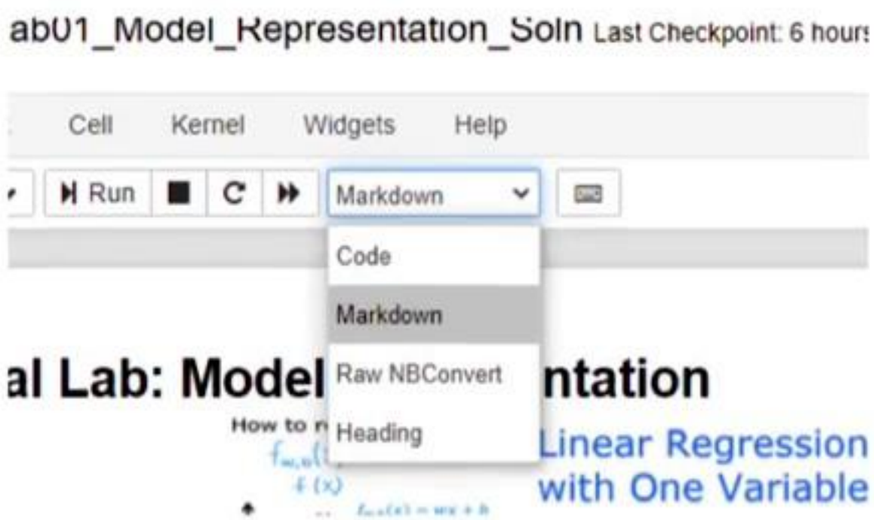


The other type of cell is the `code cell` where you will write your code:

```
In [ ]: #This is a 'Code' Cell
```

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Run Code

Linear Regression with One Variable

The other type of cell is the `code cell` where you will write your code:

In [1]:
```
#This is  a 'Code' Cell
print("This is  code cell") # <shift> + <Enter>
```

```
This is  code cell
```

## Python

You can write your code in the code cells. To run the code, select the cell and either

- hold the shift-key down and hit 'enter' or 'return'
- click the 'run' arrow above

View Insert Cell Kernel Widgets

Run Markdown

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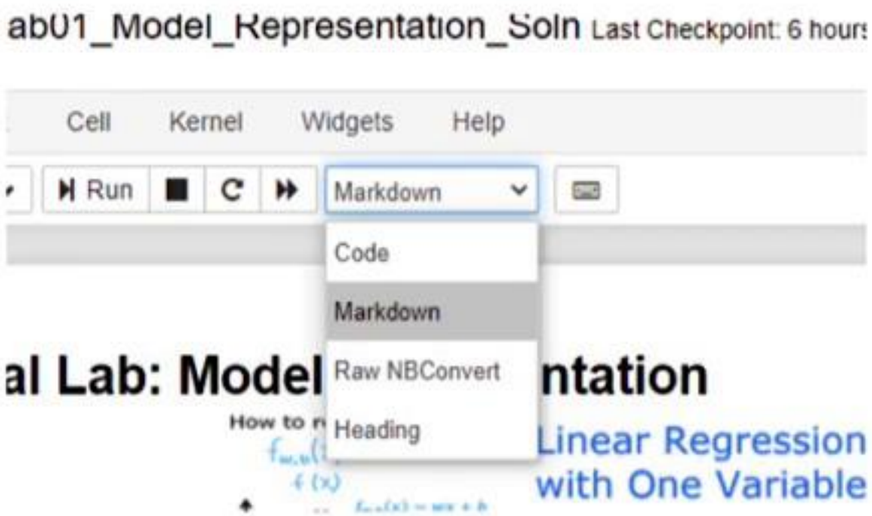
Hi, how are you? ^_^ =). ^_^



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Run Markdown

with One Variable

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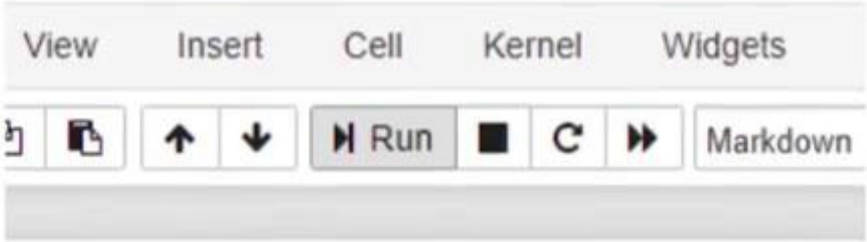


# Optional Lab: Model Repr

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Run Markdown

Run Markdown

# Optional Lab: Model Repr

### Print statement

Print statements will generally use the python f-string style.
Try creating your own print in the following cell.
Try both methods of running the cell.

In [2]:
```
# print statements
variable = "hi Andrew!"
print(f"f strings allow you to embed variables {variable}")
```

```
f strings allow you to embed variables hi Andrew!
```

## Congratulations!

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Run Markdown

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```
f strings allow you to embed variables hi Andrew!
```

## Congratulations!

You now know how to find your way around a Jupyter Notebook.

In [ ]: